

RICHER SIMULATIONS

FS GROUND POLYGON TO XP TOOL

(FSGP2XP Beta)

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Introduction

The idea of this tool is to make life easier for developers who have mostly completed their airports for FSX/P3D to use their **source** files to insert the same airport into WED for XP11.

This tool is **NOT** a conversion tool like FS2XP. You **CANNOT** take already released airports and convert them from an FSX/P3D format (i.e. MDLs, BGLs, etc.) to an XP11 format (i.e., OBJs, DSFs, etc.).

It was originally meant to convert Ground Polygons designed in SketchUp to be converted to WED Draped Orthophotos (.pols), hence the name FS GP 2 XP, but has since been expanded to create some Line Placements (.lins), Object Placements, Ramp Positions, Truck Parking Locations, Taxi Routes and Nodes, and Airport Chains and Nodes.

These are all classes within WED. Hence, the tool really makes inserting elements directly into WED easier, without all the alignment guesswork.

Tools needed for input (in this manual, I will not be explaining the use of these tools):

SketchUp aka SU (Preferred version 2018 pro, I'm not sure if the OBJ format it exports differs in other versions)

Airport Design Editor aka ADE (For Taxi Routes and Parking Placements)

ModelConverterX aka MCX (For converting your models into XP OBJs split by material)

BGLAnalyzerX (For extracting MDLs and XMLs from Placement BGLs)

The Tool: FS Ground Polygon to X-Plane

The tool is split into four sections: Ground Polygon Conversion, Parking Conversion, OBJ Placement and MDL Name Edit.

We will start with the simplest section: OBJ Placement.

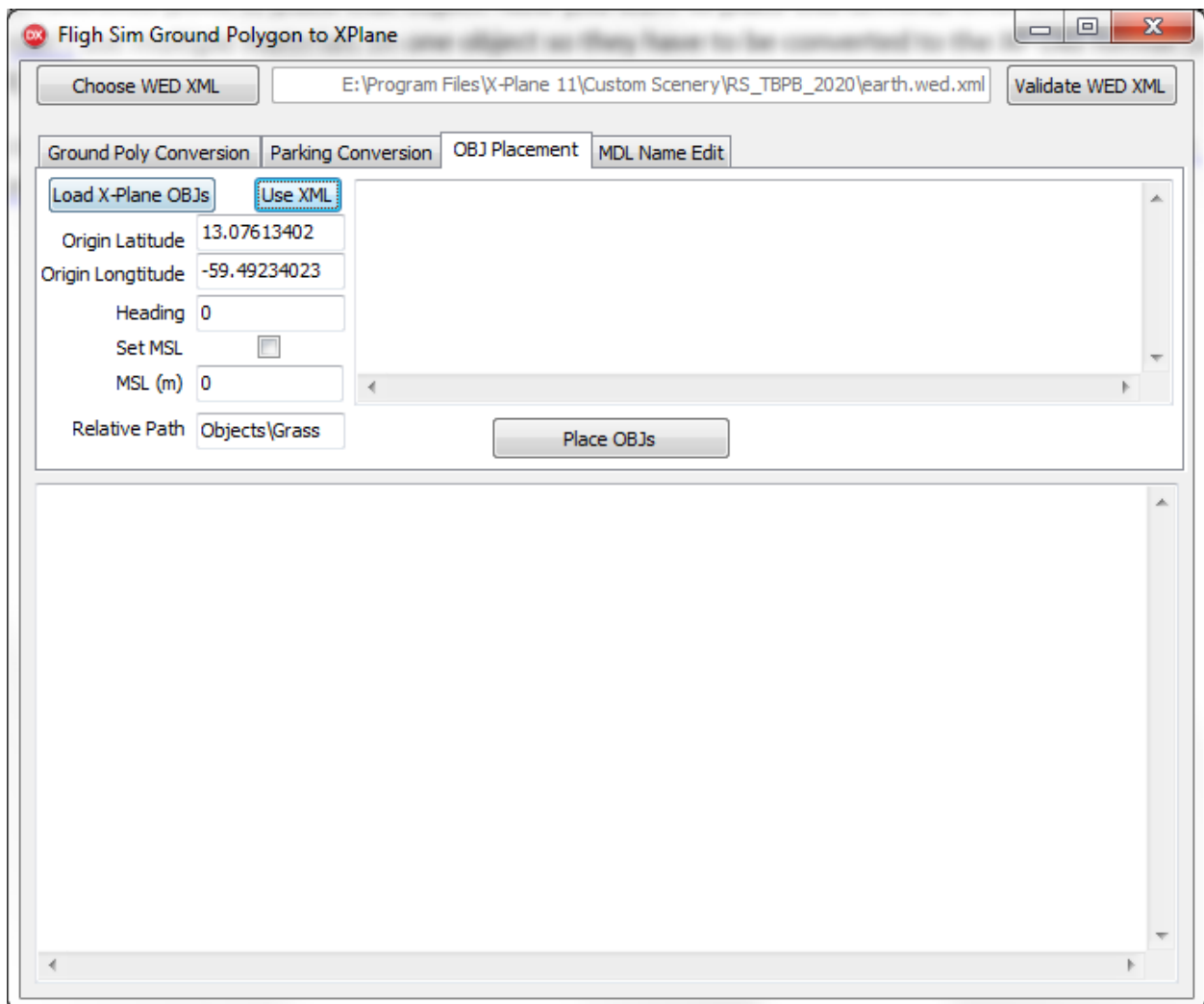
OBJ Placement

You have built your entire terminal as one model for FSX/P3D with multiple materials and probably use a single reference point to place that object. Now you want to place this terminal in XP and discover that you cannot use multiple materials on one object so they have to be converted to the XP OBJ format (using MCX) which splits it by material.

This section allows you to simply select all the resulting OBJs and use that single reference point to place them into WED automatically.

Simply:

- 1) Choose the Earth.Wed.XML file that you want to insert into.
- 2) Select Load X-Plane OBJs.
- 3) Select all the OBJs you want to place.
- 4) Enter the Origin Latitude and Longitude.
- 5) Enter the heading and altitude (MSL) settings.
- 6) Enter the relative path to the objects that you will be placing. This is relative to the scenery package folder, so if all your OBJs are in a sub folder called "My_Scenery\Objects\Terminal" enter "Objects\Terminal". If they are in the root folder, leave this blank.
- 7) Click on Place OBJs.



However, what if you don't use a single reference point and all your objects are positioned using individual latitudes and longitudes such as in a BGL placement method?

Well you can use BGLAnaylzerX to extract that positioning information into the XML, and convert the MDLs that were also extracted into OBJs, and use the XML information to place them into WED instead.

Simply:

- 1) Click Use XML.
- 2) Choose the XML that contains the positioning information of the objects you want to place into WED.
- 3) The heading/altitude information in the tool will become greyed out as all this info will be pulled from the XML.
- 4) Enter the relative path to the objects that you will be placing.
- 5) Click on Place OBJs.

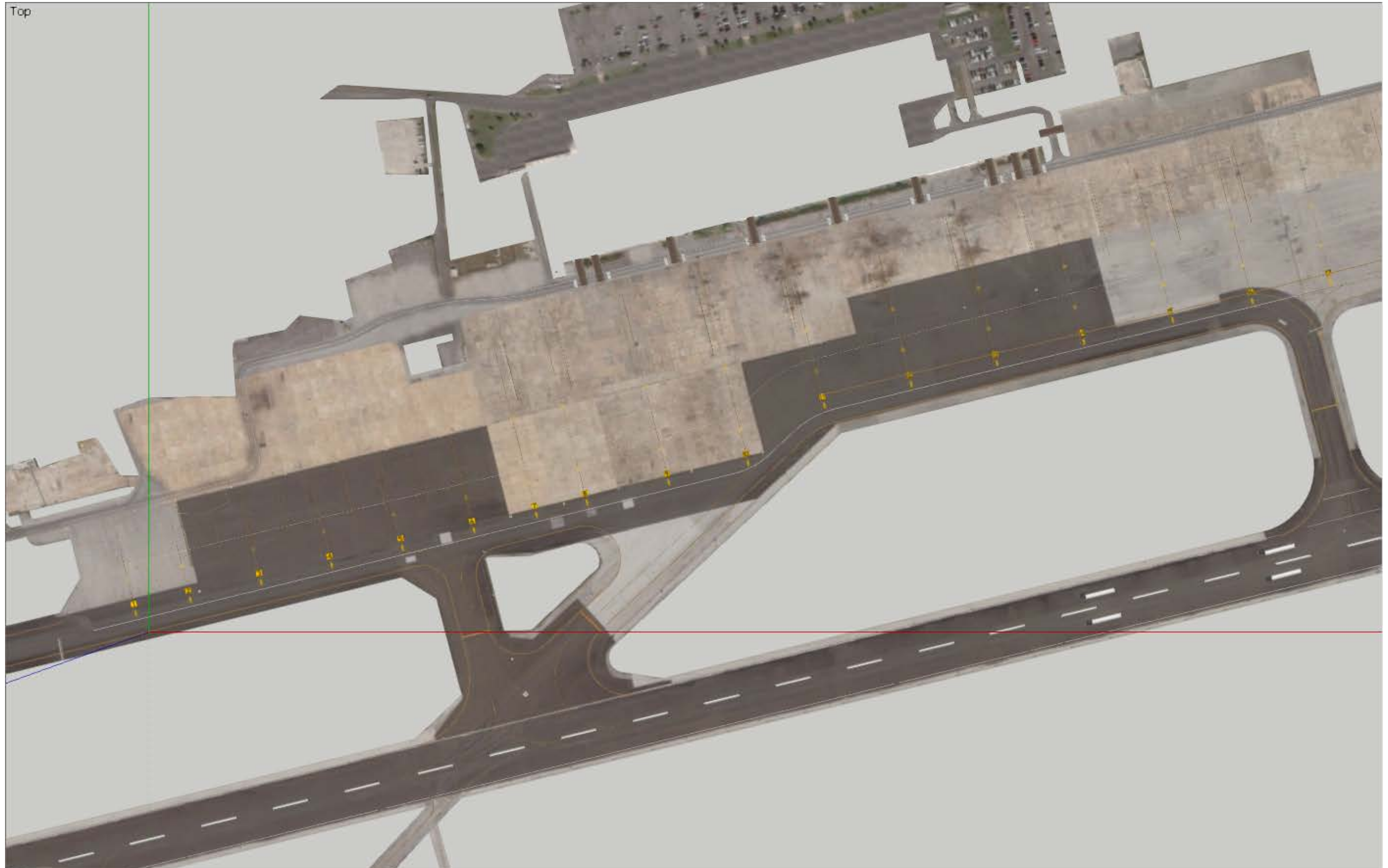
Now you can open WED and check that everything is in the right place.

Ground Polygon Conversion

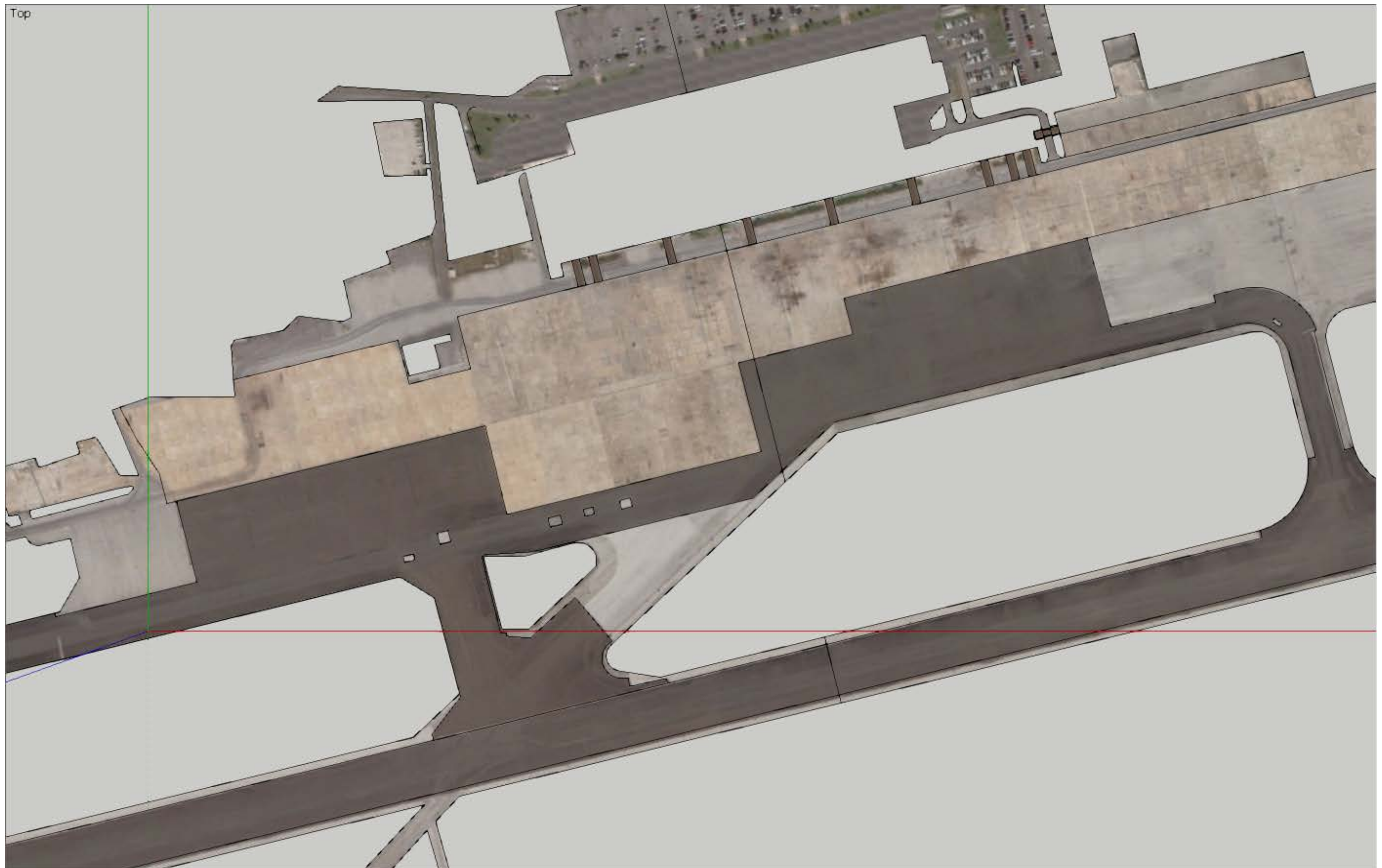
Processing Polygons

Most FS Ground Polygons can be placed in XP as draped objects. However, you will face issues such as alignment, since the distortion of the objects gets greater the larger they are, as well as the inability to add detail textures (detail maps in FS or decals in XP) if the ground polygon you use is based off a photoreal image.

By converting the original SU polygons directly to draped orthophotos, we can eliminate both these issues. Below is an example of a section of a GP fully textured in SU.

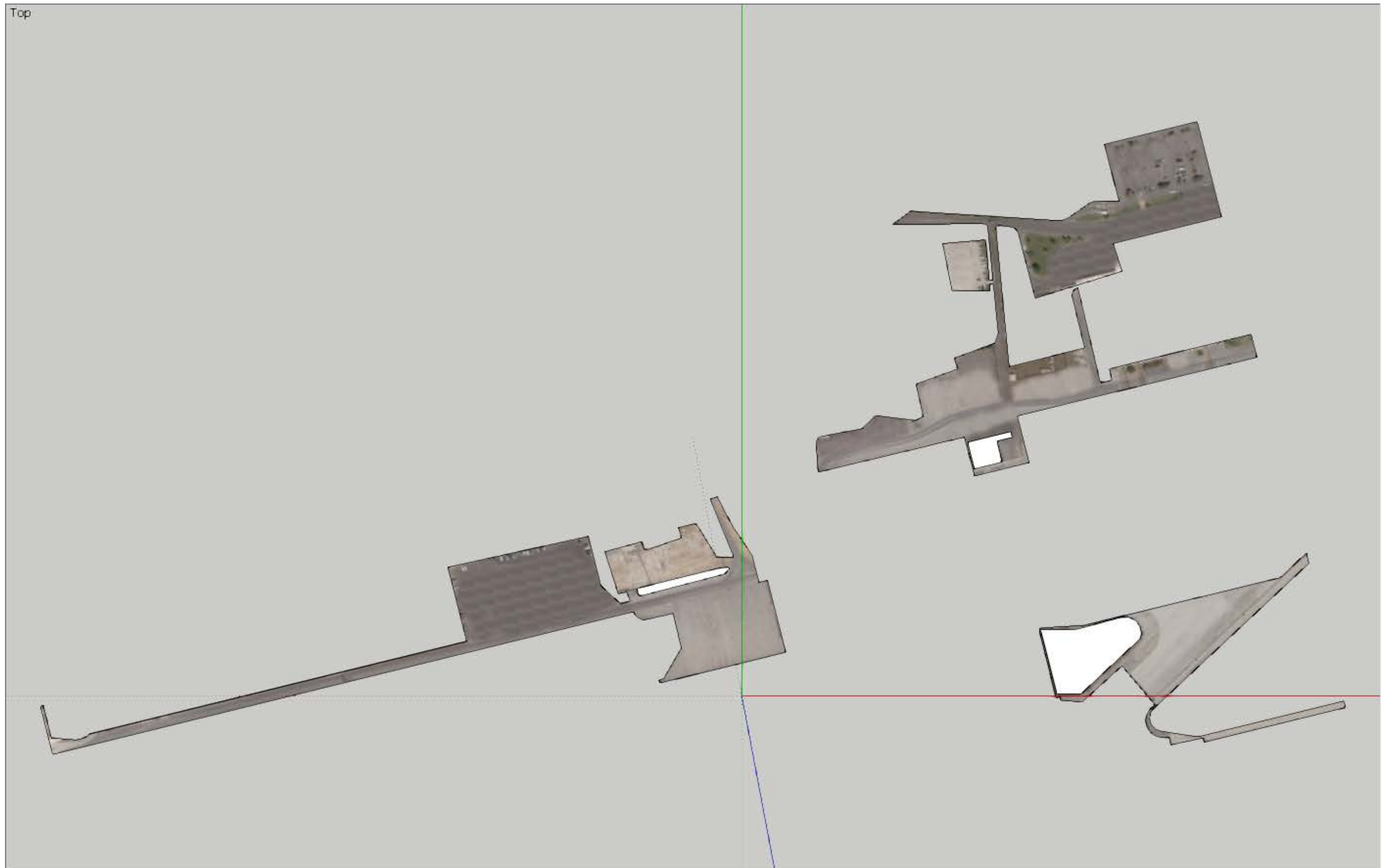


And here is the base photoreal layer that we will be converting to draped orthophotos that we can later add detail to. It is separated into sections, which will have different decals (details) such as asphalt, concrete, or tile applied to it.

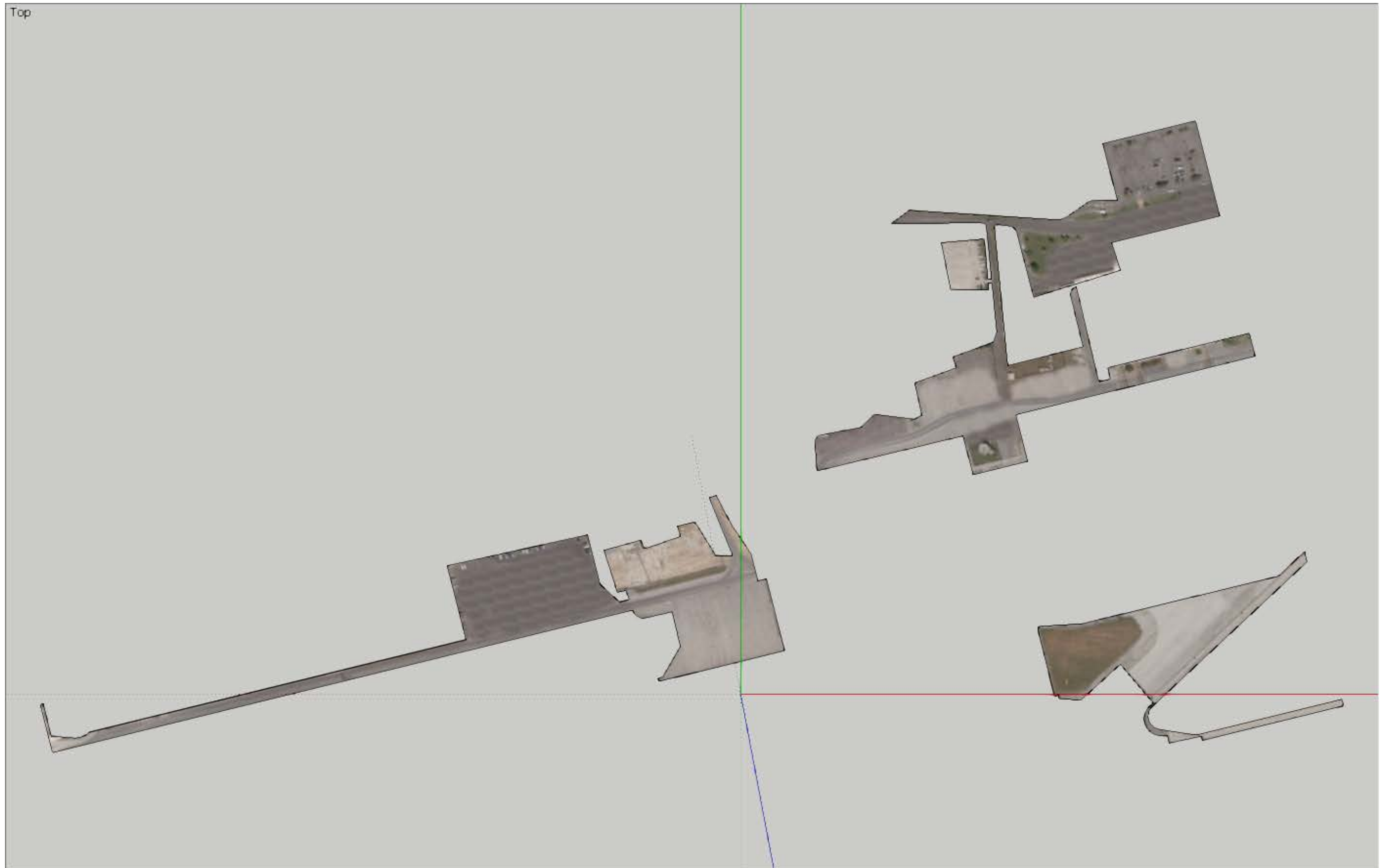


For the purpose of simplicity, let us only deal with the concrete sections. We will export these first.

You will notice that there were a few holes in the completed GP. I have filled them in here and left them as white so that they are easily visible. This will be very important later.



With all holes removed, the actual export will look like this:

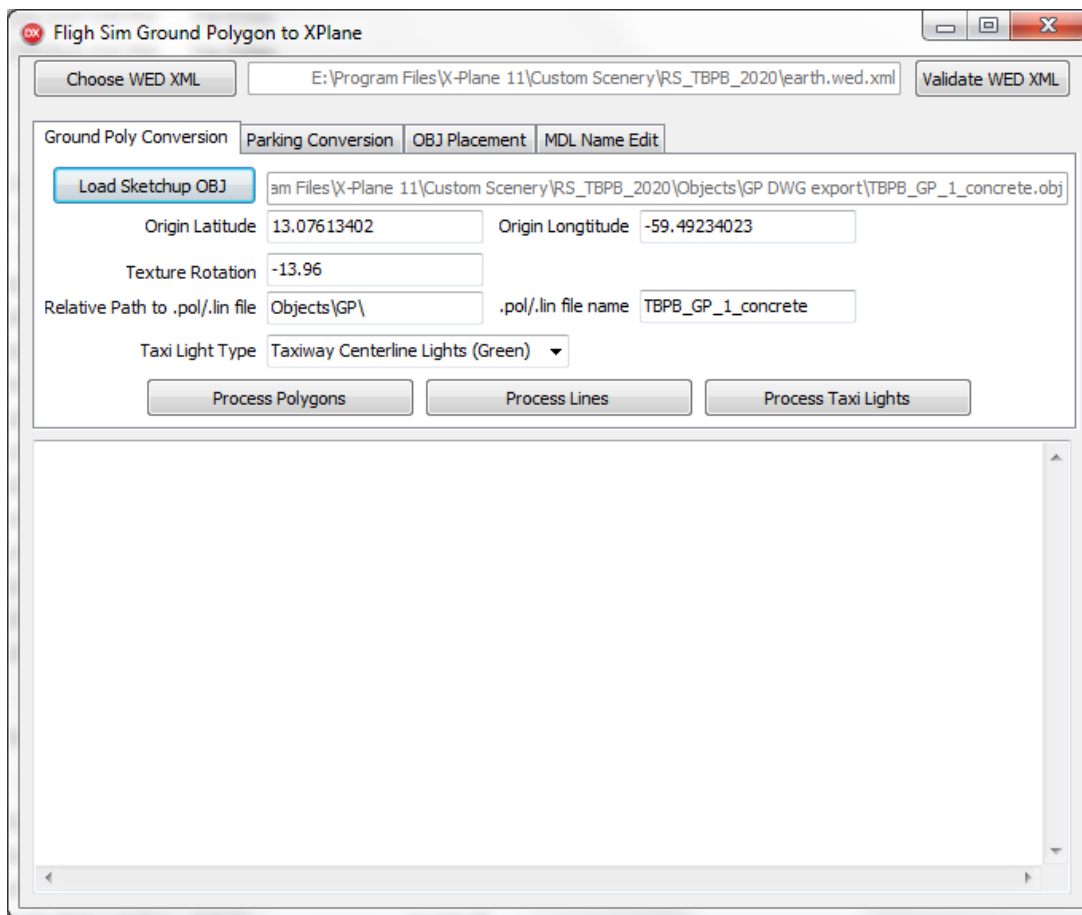


The Ground Polygon Conversion tool works with the following assumptions:

- The input .obj file is made from SketchUp
- Every mesh in the .obj file represents one complete face with NO HOLES.
- The faces are NOT triangulated.
- Holes are represented as SEPARATE FACES with INVERTED NORMALS which are in the same MESH GROUP as the face it will be a hole in. (I will explain this more later).

When you have ensured that your Ground Polygon has no actual holes, is not triangulated and is fully textured, the process of bringing them into WED as textured draped orthophotos is very simple.

- 1) Export all the GP sections that share the same texture, example concrete. I usually export the SketchUp .obj using the same name as the .pol that it will use. You most likely have already created this .pol (or at least you should have 😊)
- 2) In FSG2XP, choose the Earth.Wed.XML file that you want to insert into.
- 3) Select Load SketchUp OBJ.
- 4) Enter the latitude and longitude of the origin point of the entire SU model.
- 5) If you had rotated your texture as I normally do for better alignment to runways and taxiways you can enter it in Texture Rotation. Otherwise, leave it as zero.
- 6) You will notice that the .pol file name changes every time you load a new SketchUp .obj (to automatically match step one). You can change this if you like.
- 8) Enter the relative path to .pol. This is relative to the scenery package folder, so if all your .pol files are in a sub folder called "My_Scenery\Objects\GP" enter "Objects\GP". If they are in the root folder, leave this blank.
- 9) Ignore Taxi Light Type. This will not be used when processing Ground Polygons.
- 10) Click on Process Polygons.

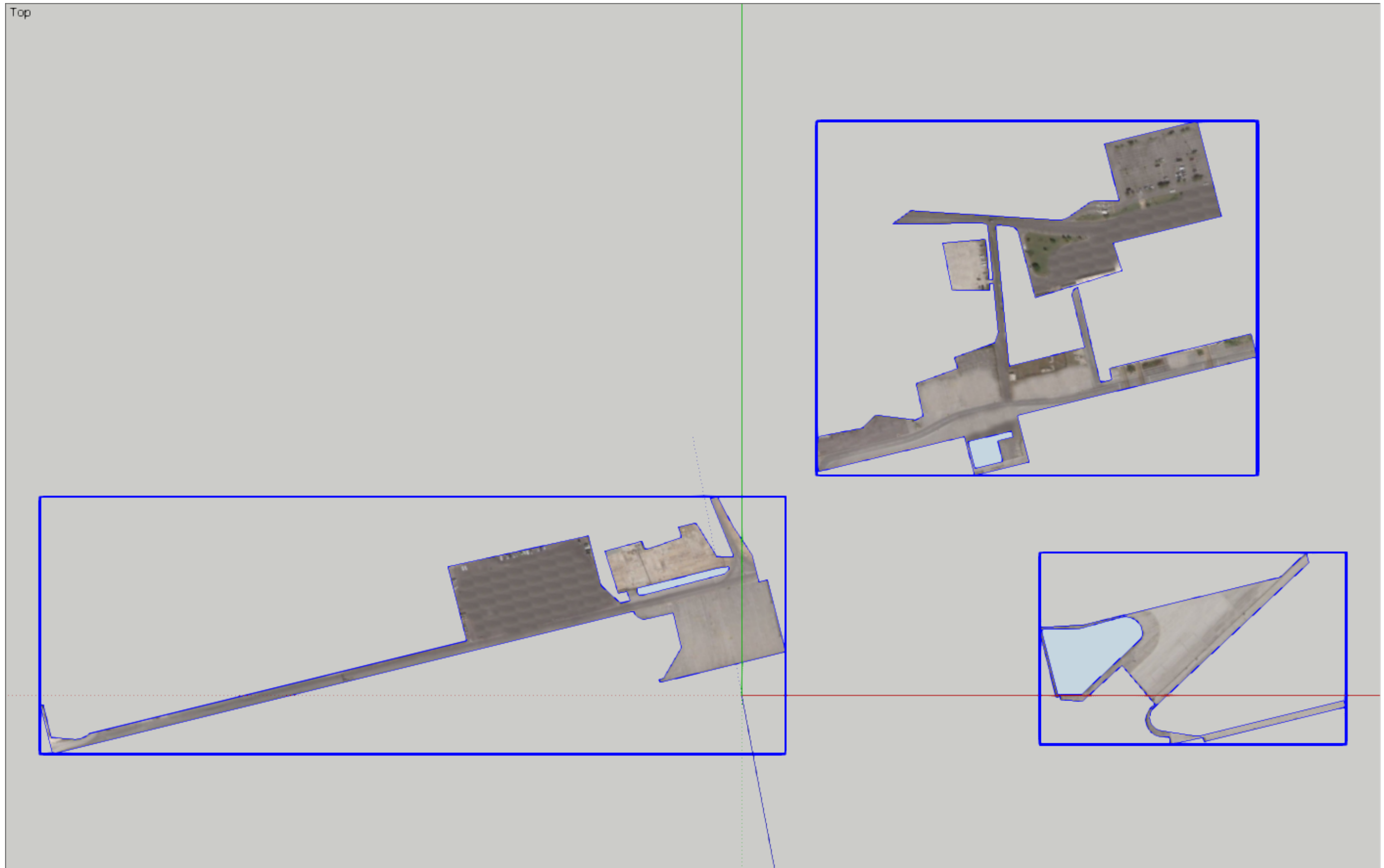


Dealing with Holes

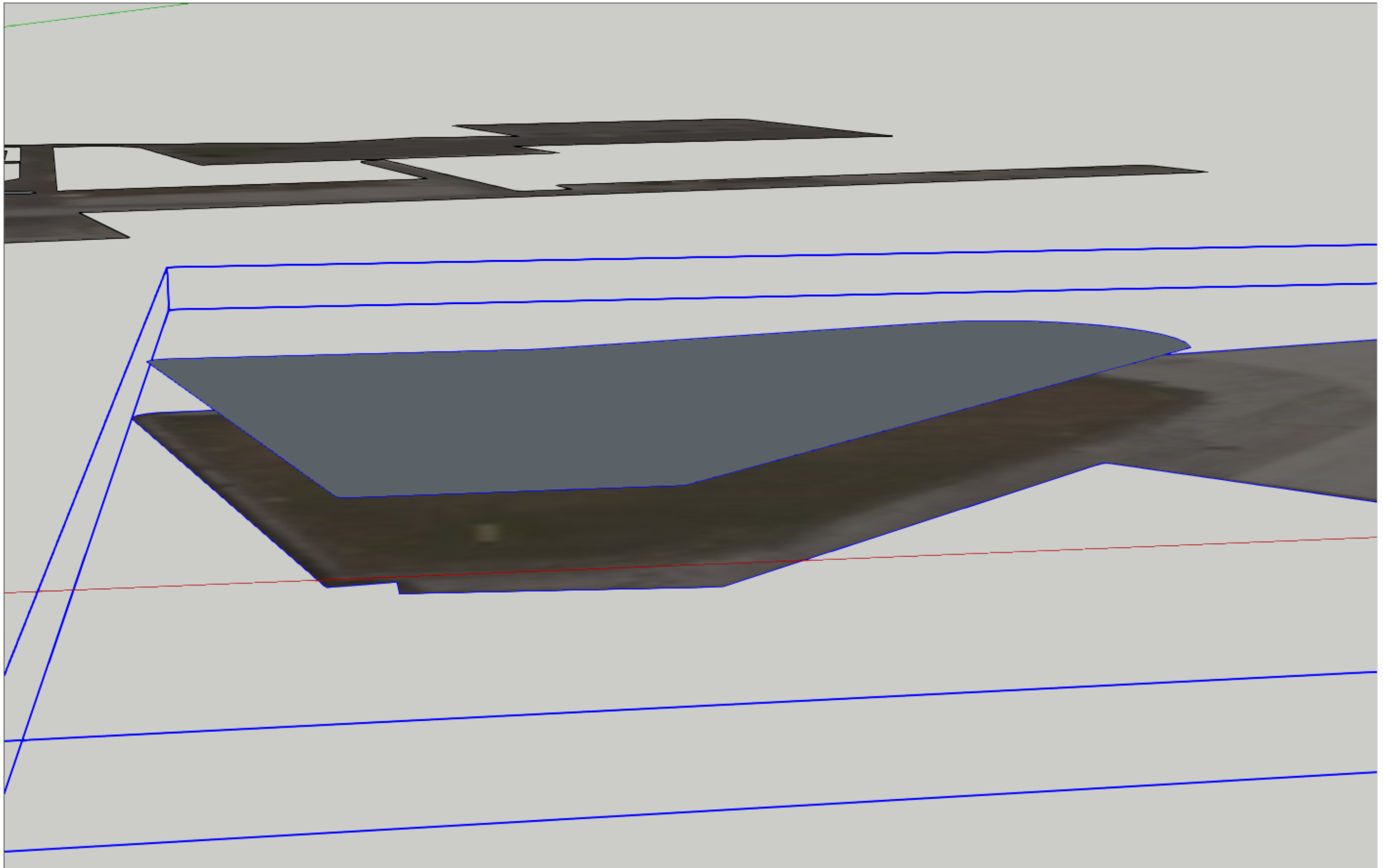
Earlier I mentioned that, in your SketchUp Ground Polygon:

- Holes are represented as SEPARATE FACES with INVERTED NORMALS that are in the same MESH GROUP as the face it will be a hole in.

This is what it should look like: You will notice that the white faces that represented the HOLES are now REVERSED, and that each of these are now part of a MESH GROUP along with the face they were originally a hole in.



However these faces are still SEPARATE from the textured face itself; they are slightly raised.



This ensures that FSG2XP can identify that the face with the INVERTED NORMAL is actually a hole in the textured face directly below it when it is exported via SketchUp.

Now you can follow the same process as above and get draped orthophotos with holes in the correct places in WED.

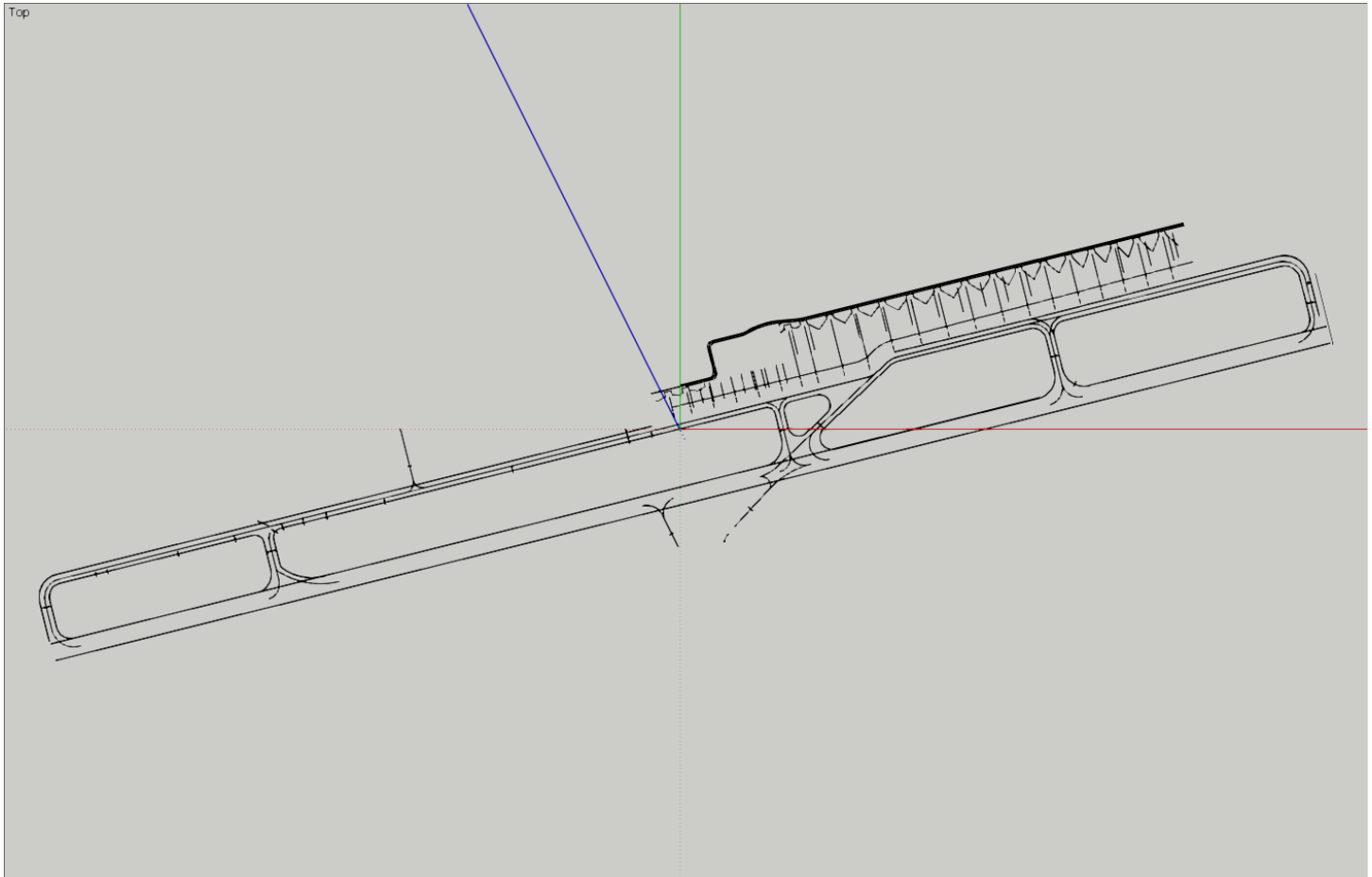
Now when opening your WED project you should have textured draped orthophotos in their correct positions (with holes ☺).



Processing Lines

Creating textured lines (such as taxi lines or Ground Polygon edges) is simplified as you most likely already created these lines in SU or they exist at the edges of your polygons. You can use the fact that SU can export lines in its OBJ format to your advantage.

Here I have the lines that represent most of the markings on my GP:



I already have a .lin file that represents all my double lined yellow taxi lines, so I can now:

- 1) Select these lines in SU, enable the export of lines within SUs export options and export an OBJ.
- 2) Choose the Earth.Wed.XML
- 3) Select Load SketchUp OBJ.
- 4) Enter the latitude and longitude of the origin point of the entire SU model.
- 5) Ignore Texture Rotation.
- 6) Change .lin file name if you need to.
- 7) Enter the relative path to the .lin.
- 8) Ignore Taxi Light Type.
- 9) Click on Process Lines.

Note:

FSG2XP will attempt to automatically:

- Close lines that form a loop when inserting them into WED.
- Separate lines that share common vertices.

Line insertion can get very messy which can cause WED to crash and/or not save WED files and/or cause WED XMLs to become corrupt all together. It is recommended to keep line intersection in SU to an absolute minimum and to test your WED file every time you process a new set of lines into the WED XML. FSG2XP saves a backup copy of your WED XML after every process, just in case.

Processing Taxi Lights

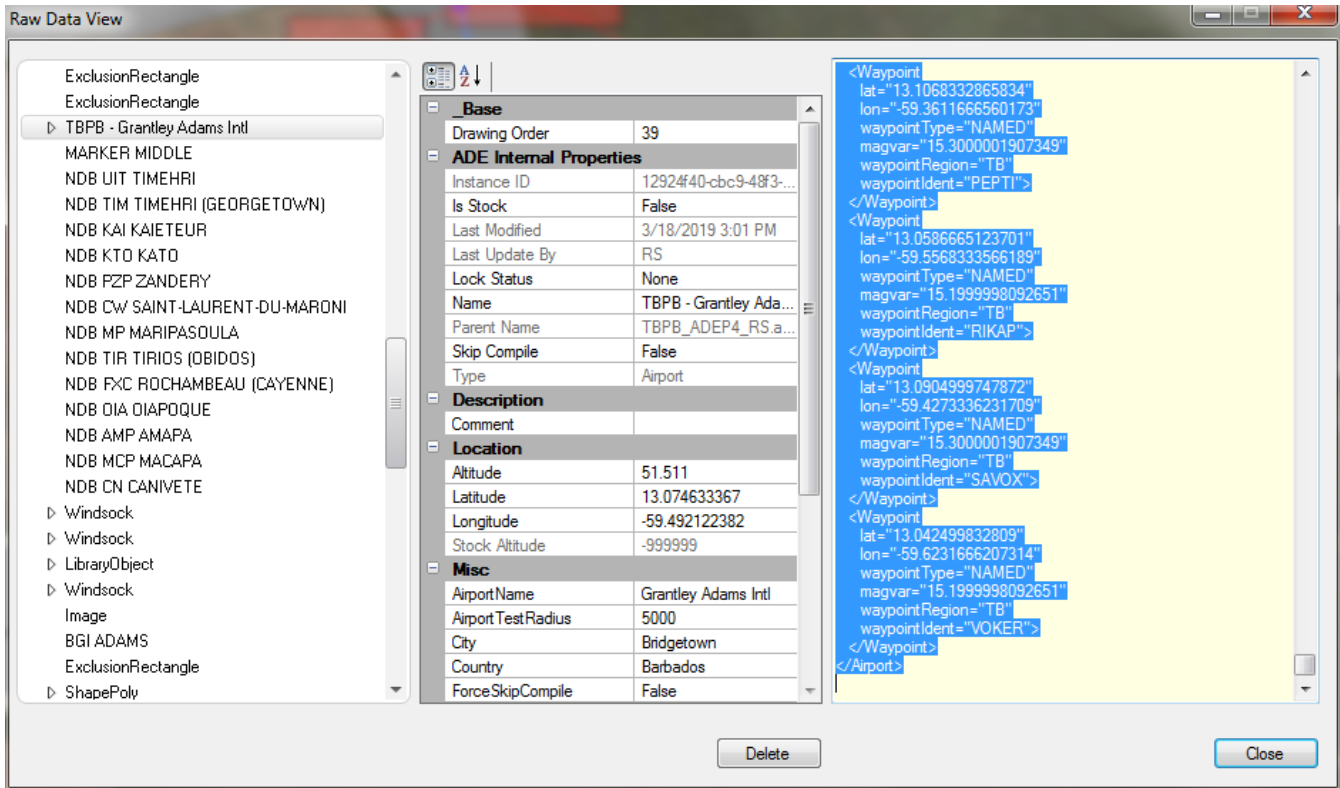
The process for creating a string of Taxi Lights is the same as processing lines above. Except you ignore the texture and .lin settings and then click on Process Taxi Lights. For now, only Taxiway Centerline Lights (Green) can be selected from the drop down menu, but this can easily be changed from within WED. If after choosing a new type of light from within WED they appear on the wrong side of the line, simply reverse the line direction using CTRL+SHIFT+R.

Parking Conversion

You have already aligned your parking routes and spaces to perfection in Airport Design Editor. Why do it all again in WED?

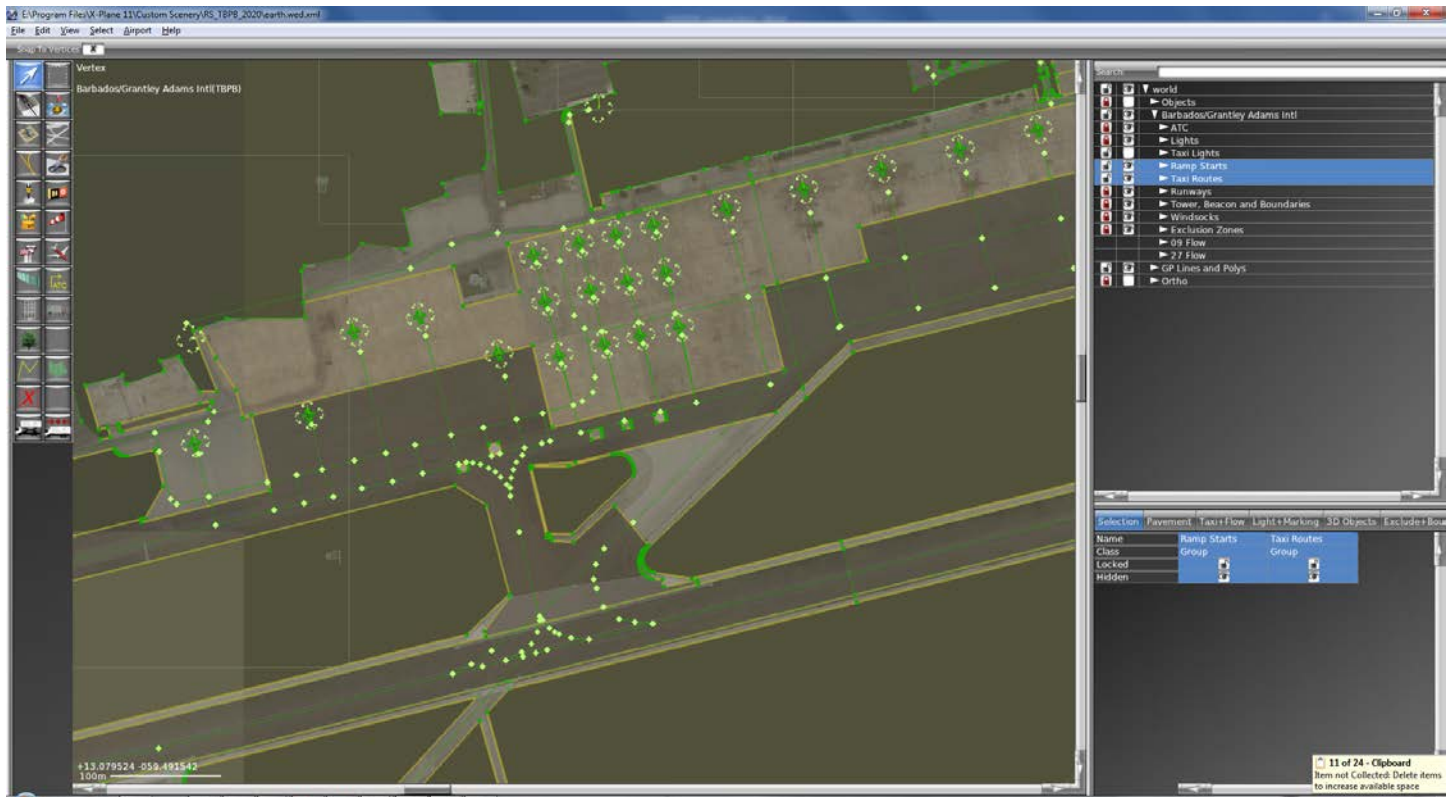
Thankfully with a few tweaks, FSGP2XP can read ADEs data and convert it to a WED Format.

- 1) Open your project in ADE and go to Tools > Raw Data View
- 2) Expand the left side and scroll down to the name of your airport. The XML code will appear on the right hand side.
- 3) Select all the text and copy and save it into a text XML file.



- 4) In FSG2XP select this file and click on Process Parking Spaces and Taxi Paths.

The imported Taxi Ramps and Routes in WED:



Note:

- WED does NOT allow crossing taxi paths. There must be taxi points to merge the intersecting paths instead of the paths crossing over one another.
- Some paths cannot intersect with a runway, example vehicle paths. You will have to modify your routes to account for this.
- XP places user/AI aircraft nose wheels exactly on the taxi parking space point whereas FSX/P3D places the aircraft center at this point. You may manually have to move up all the parking points slightly to account for this.
- FSG2XP will automatically attempt to set the parking Ramp Start Type, Equipment Type, Size, Ramp Operation Type and Airlines from the XML data. Some additional tweaking may be necessary in WED.
 - Suitable aircraft size is determined from parking space radius
 - Fuel spots are deleted
 - Vehicle parking spots are converted to a random Truck Parking Location

MDL Name Edit

This section of the tool allows for batch editing of the internal Friendly Name of multiple MDLs that have identical Friendly Names but different File Names.

This can happen if all your MDLs were extracted from the same BGL (example: Objects.BGL) using BGLAnalyzerX. They will have separate filenames, example: Objects_1.mdl, Objects_2.mdl, Objects_3.mdl, but will all have the identical Friendly Name "Objects".

You may most likely use this when you want to, for example, batch convert multiple MDLs to the XP OBJ format in MCX without them being overwritten by each other. When performing a batch convert in MCX, it saves the file with the friendly name as the filename instead of keeping the old filename. Hence, a batch convert to OBJ will be impossible.

Simply load the extracted MDLs up in FSGP2XP and click on Edit Names and the internal Friendly Names will be changed to match the individual filenames.